Practical 6

Aim : Implement Smart contact and run it using application tool.

Smart contracts:

A **smart contract** is a self-executing computer program stored on a blockchain. It contains predefined rules and conditions, and automatically performs actions when those conditions are satisfied. Smart contracts eliminate the need for intermediaries and provide **transparency, security, and automation**.

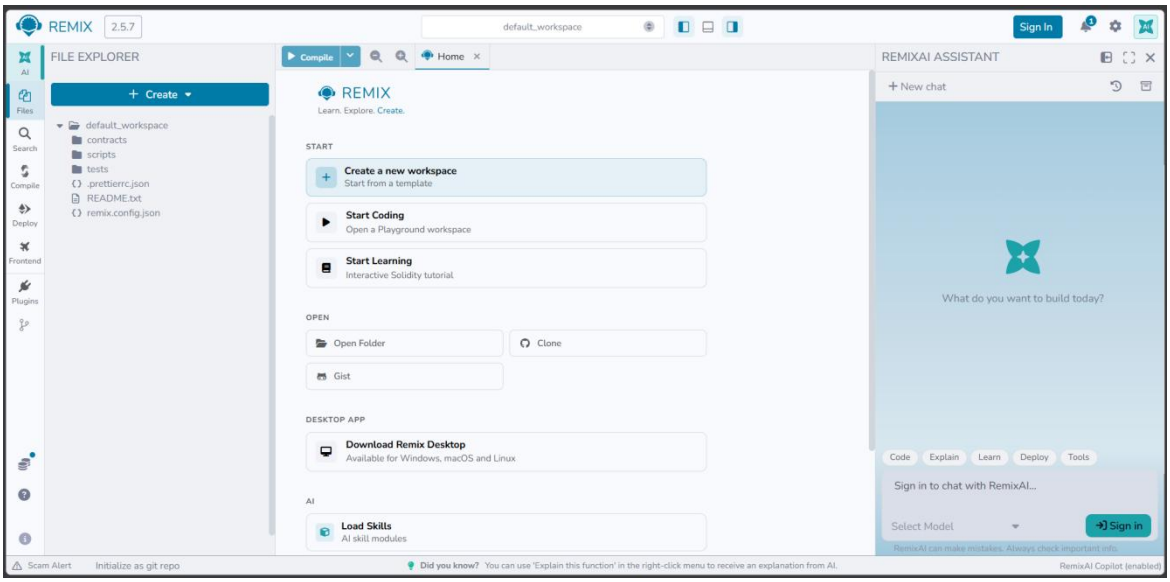
Smart contracts are commonly written using programming languages such as **Solidity** and are used in applications like cryptocurrency transactions, supply chain management, banking, voting systems, and digital asset management. Once deployed on a blockchain, the contract operates according to its programmed logic and its records are difficult to alter.

Steps :

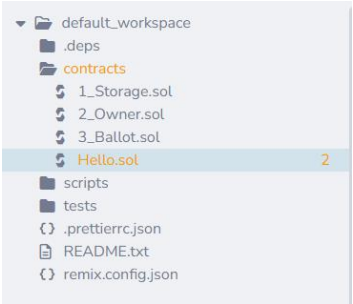
Step 1: Open Remix IDE

Open the Remix website:

https://app.remix.live/



Step 2: Create a Solidity File



Step 3: Write the Smart Contract

Enter code:

Compile | Home | Hello.sol

Create a DApp with your contracts Start now

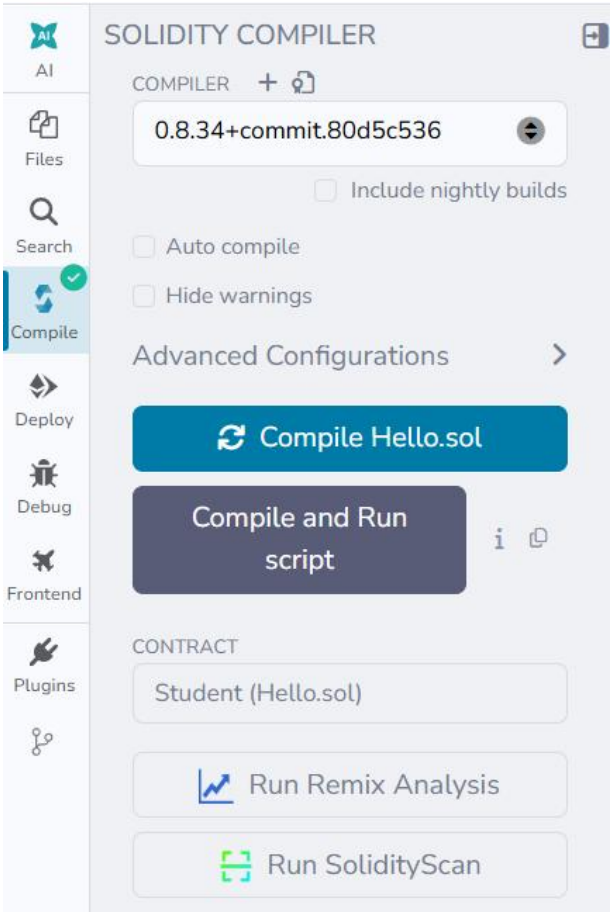
```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;

contract Student {
    string public name;
    uint public age;

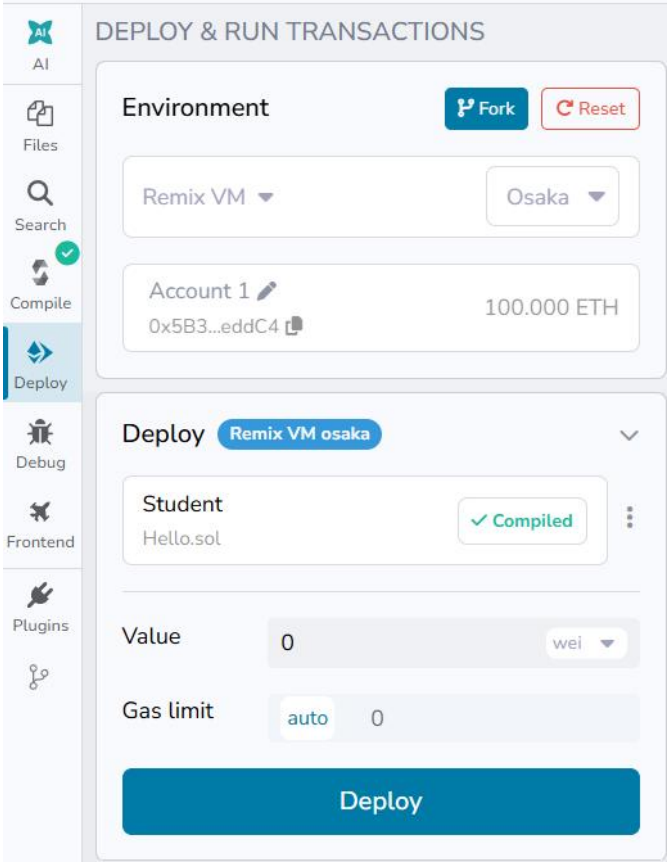
    function setStudent(string memory _name, uint _age) public {    infinite gas
        name = _name;
        age = _age;
    }

    function getStudent() public view returns (string memory, uint) {    infinite gas
        return (name, age);
    }
}
```

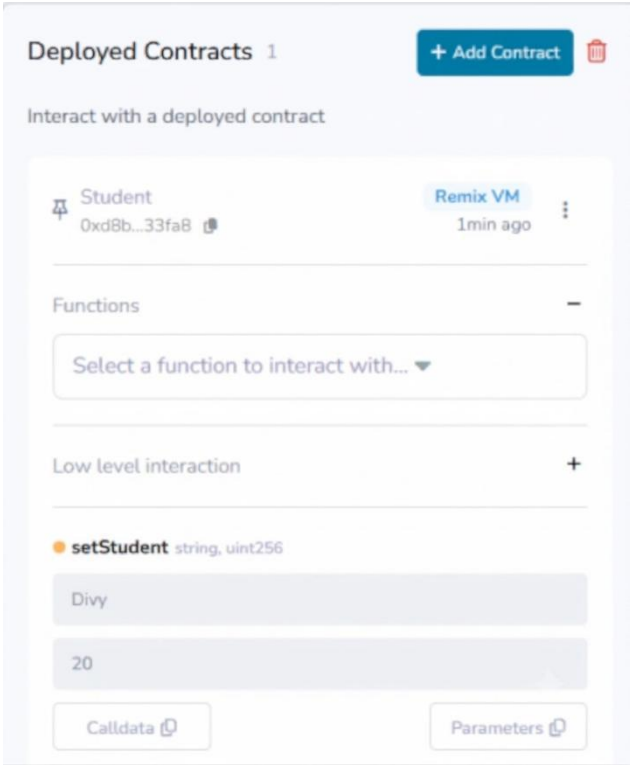
Step 4: Compile the Contract



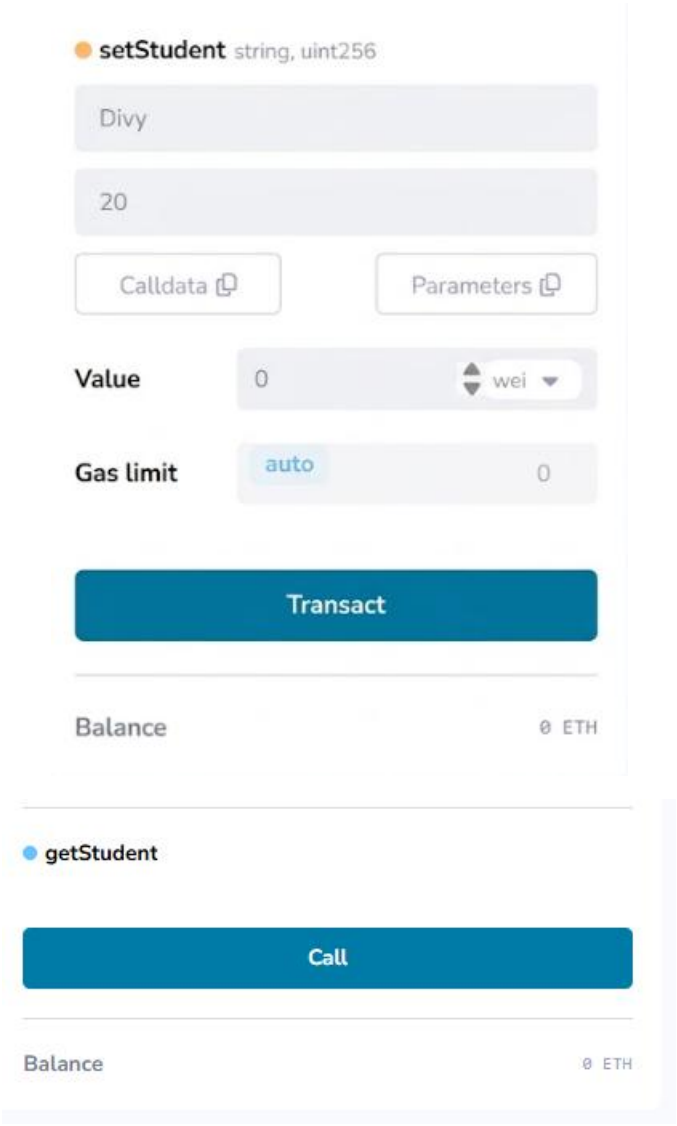
Step 5: Deploy the Smart Contract



Step 6: Run the Smart Contract



Click Transact



Click Call

